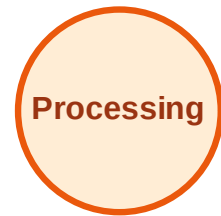


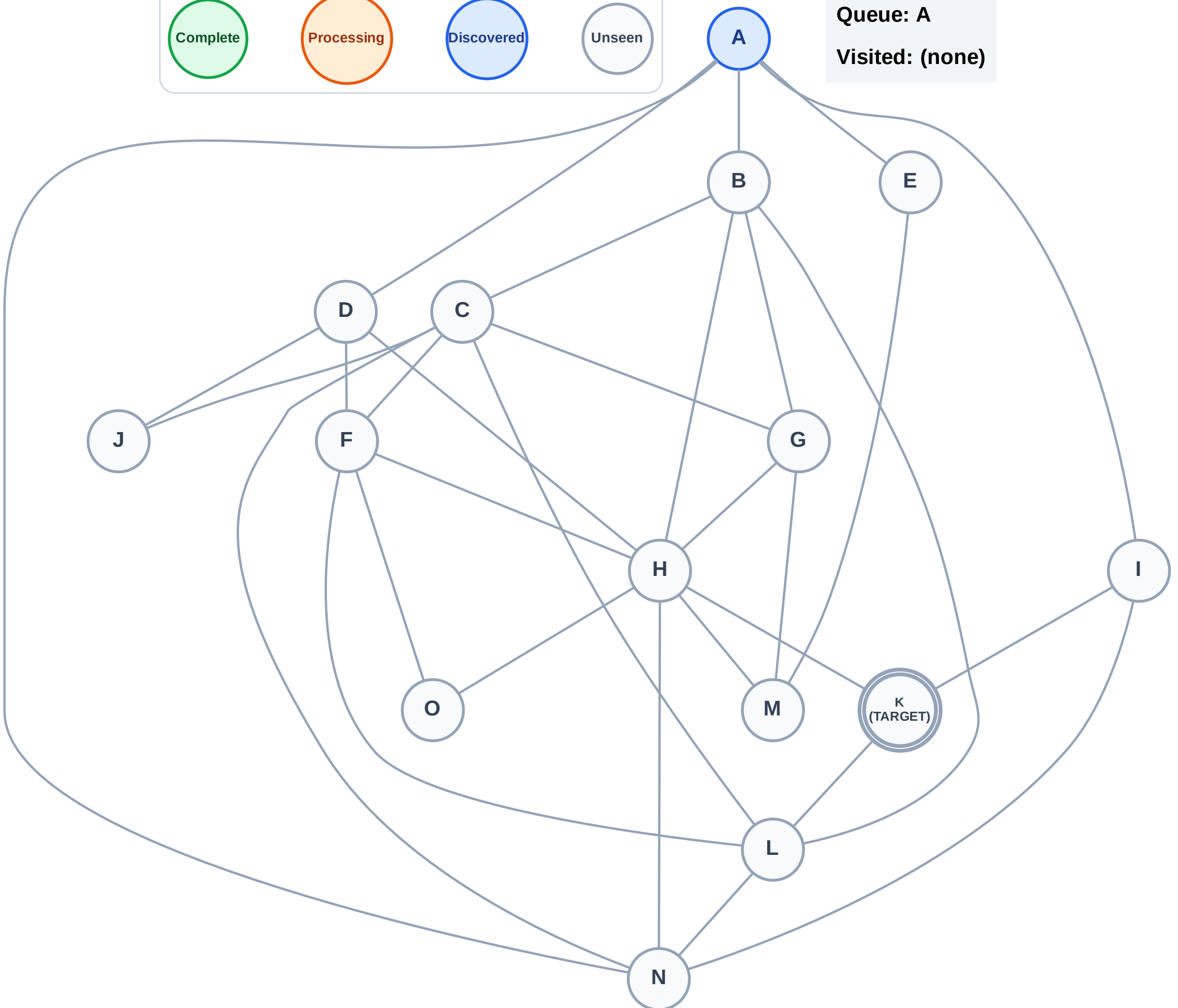
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



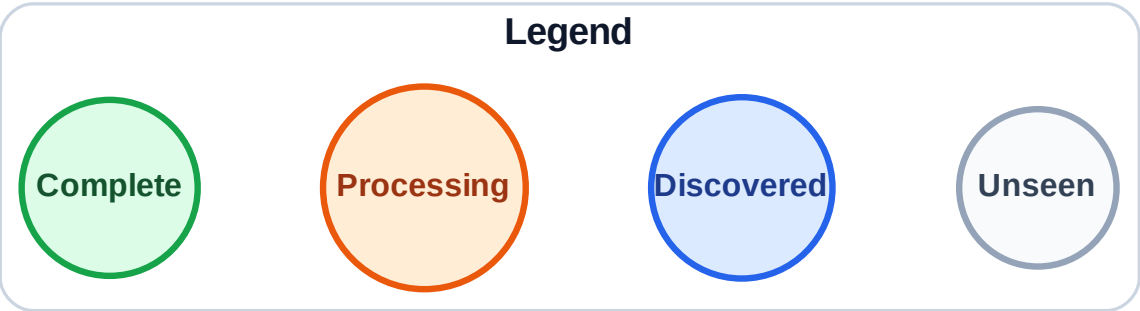
Queue: A
Visited: (none)



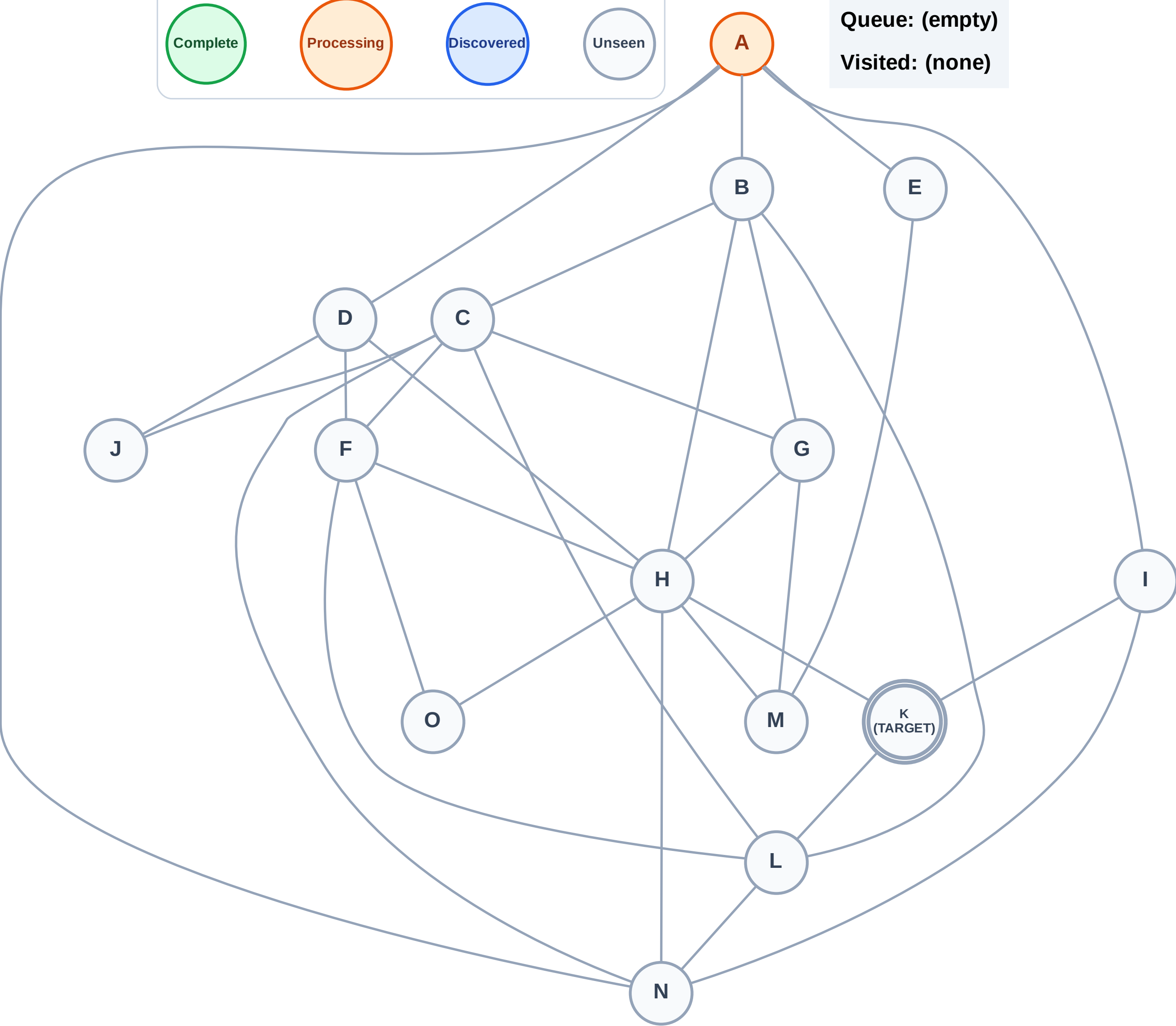
BFS Traversal

Step 2: Process node A

Legend



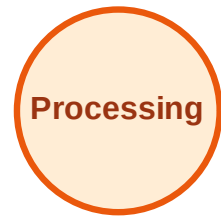
Queue: (empty)
Visited: (none)



BFS Traversal

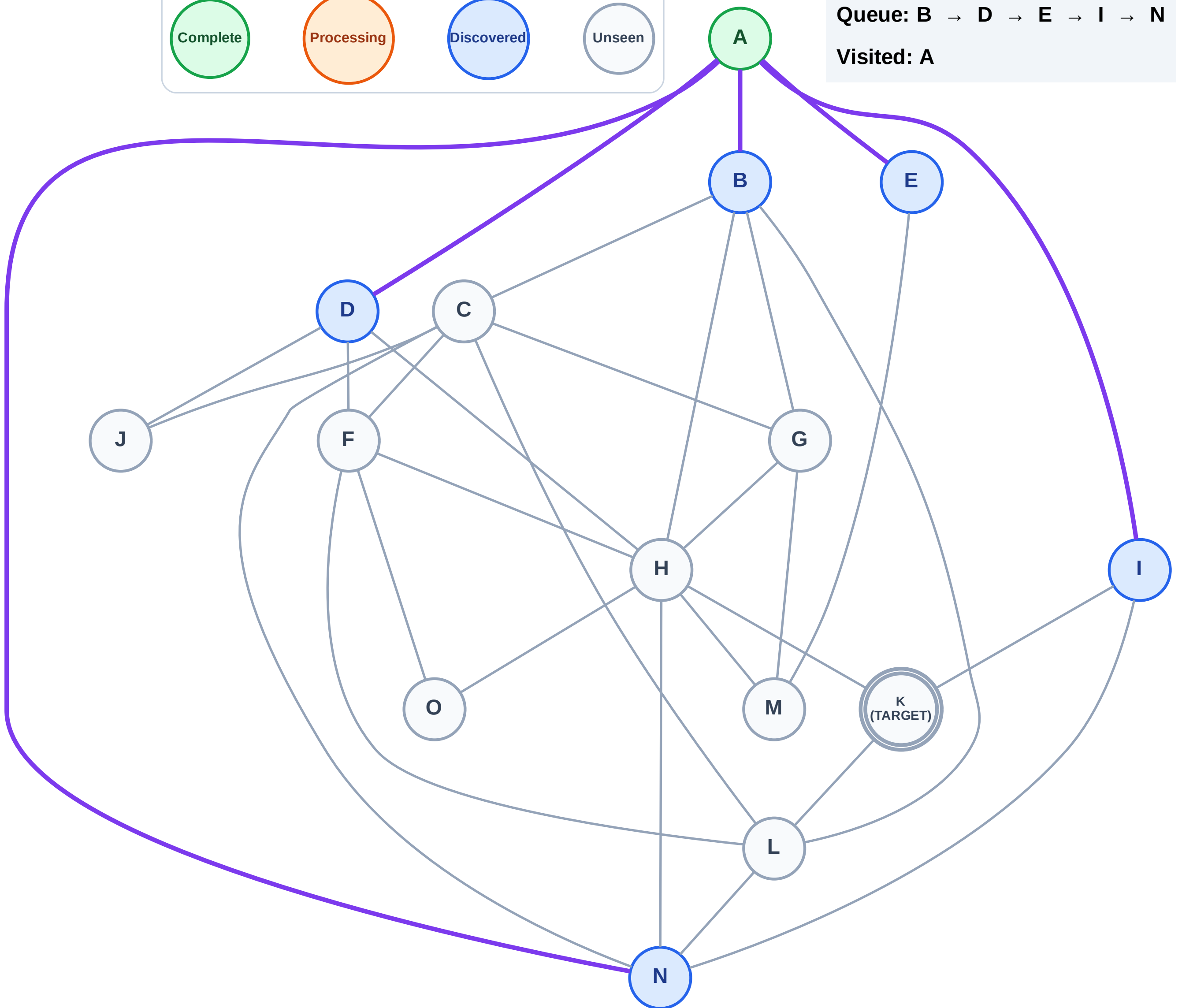
Step 3: Discover B, D, E, I, N from A

Legend

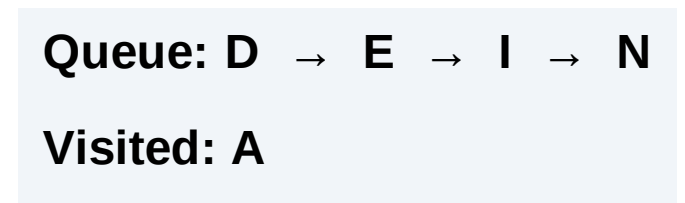


Queue: B → D → E → I → N

Visited: A



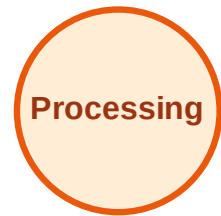
Step 4: Process node B



BFS Traversal

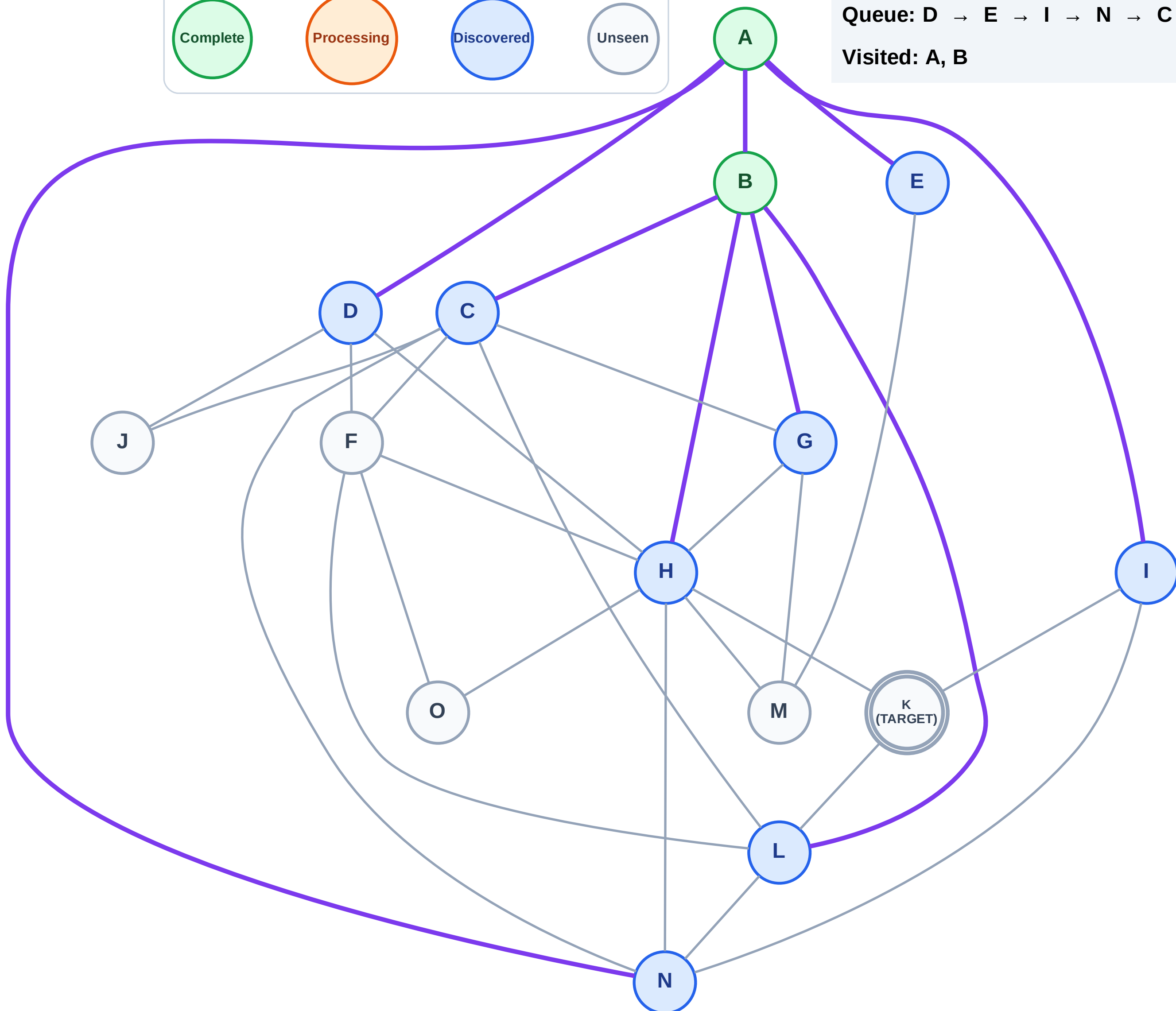
Step 5: Discover C, G, H, L from B

Legend



Queue: D → E → I → N → C → G → H → L

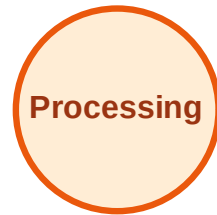
Visited: A, B



BFS Traversal

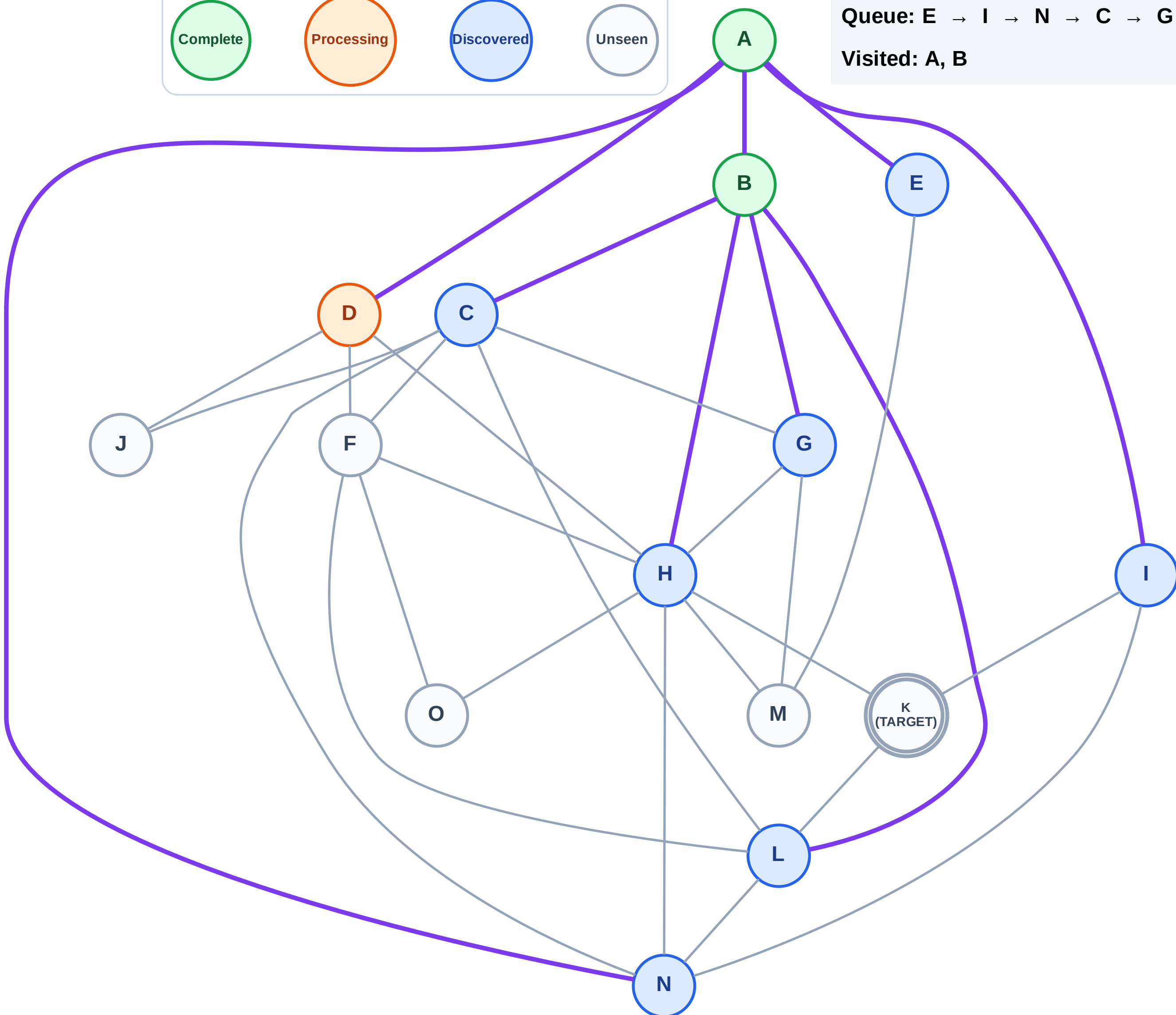
Step 6: Process node D

Legend



Queue: E → I → N → C → G → H → L

Visited: A, B



BFS Traversal

Step 7: Discover F, J from D

Legend

Complete

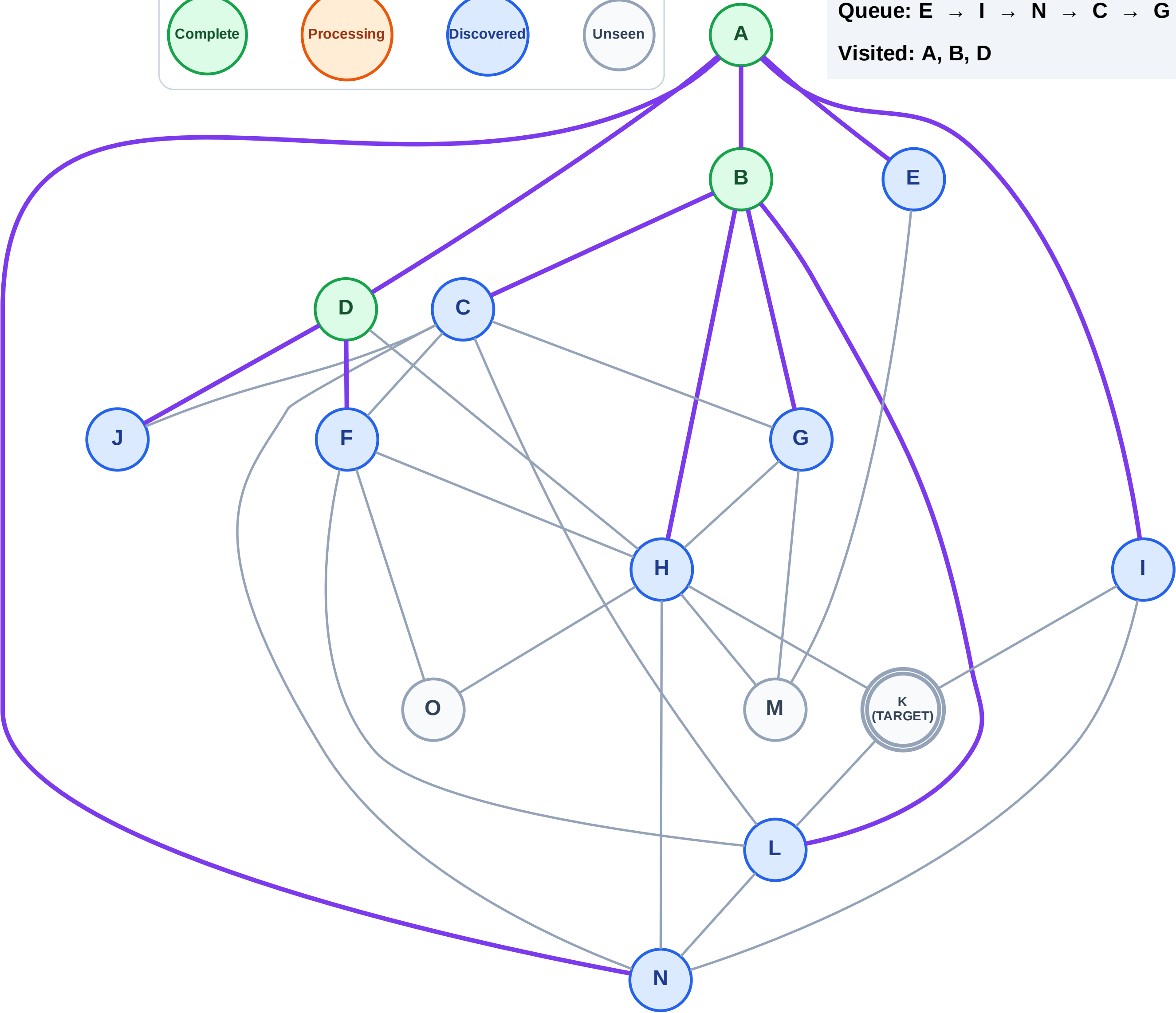
Processing

Discovered

Unseen

Queue: E → I → N → C → G → H → L → F → J

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

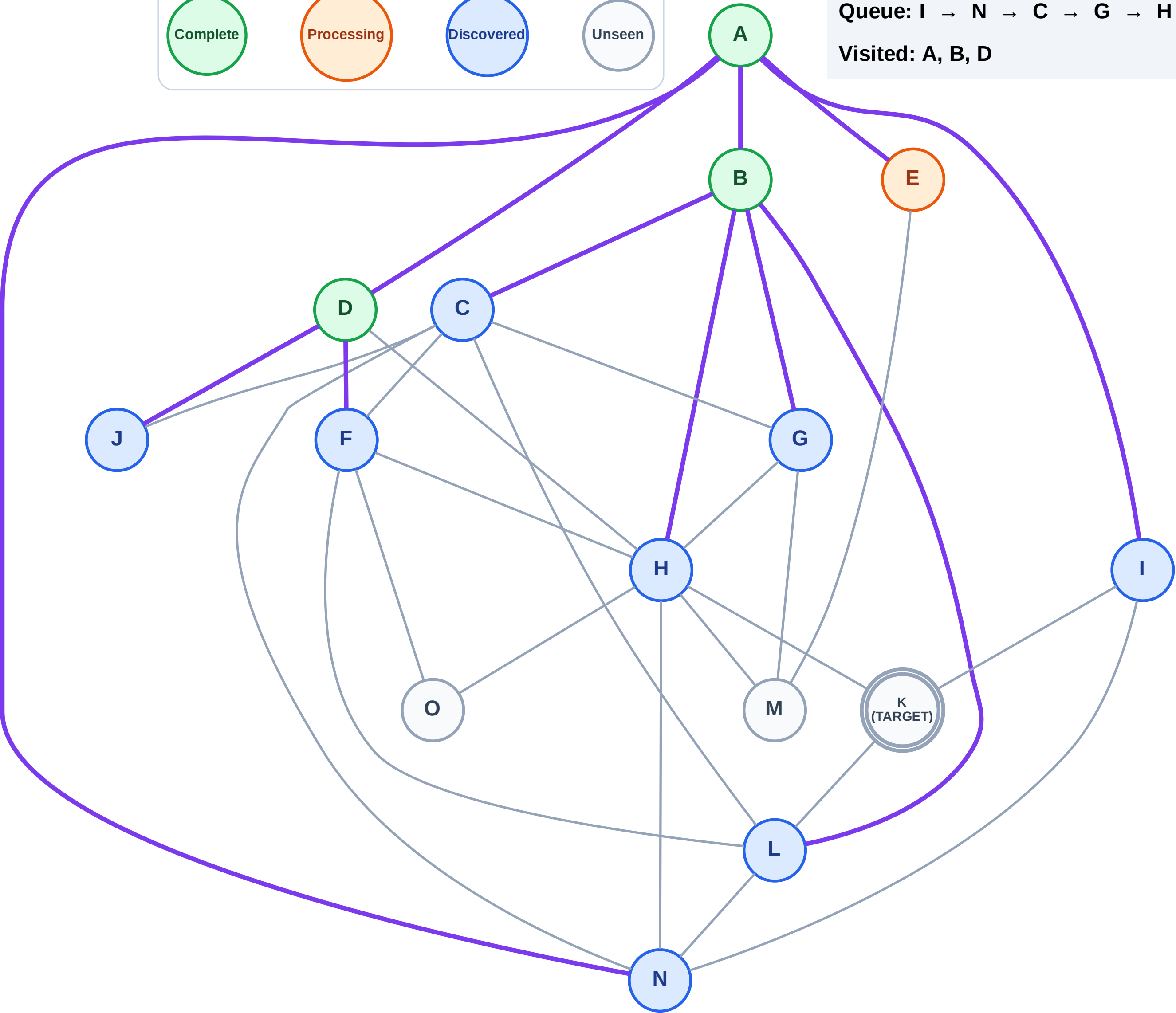
Processing

Discovered

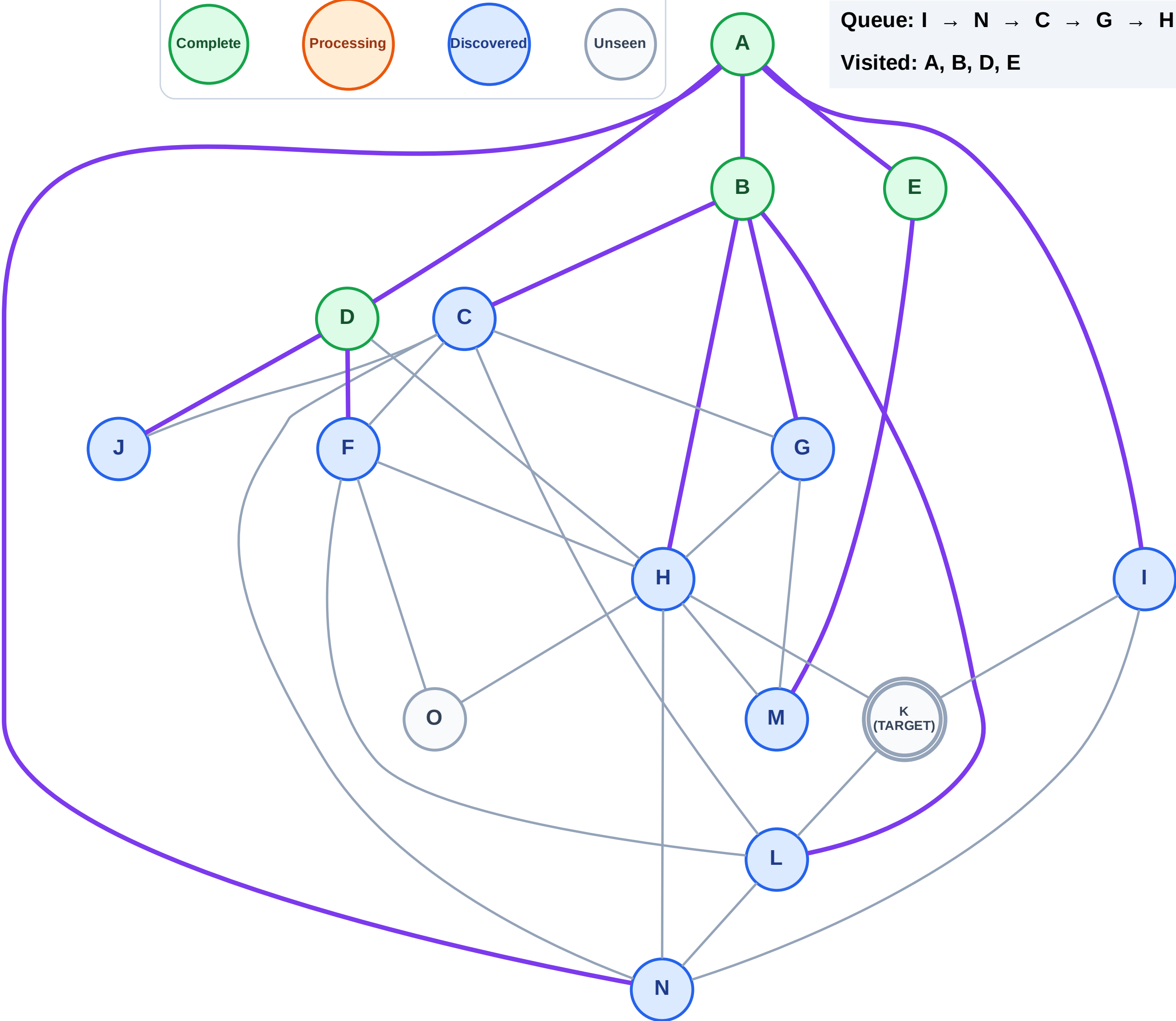
Unseen

Queue: I → N → C → G → H → L → F → J

Visited: A, B, D



Step 9: Discover M from E



Step 10: Process node I

Step 10: Process node I

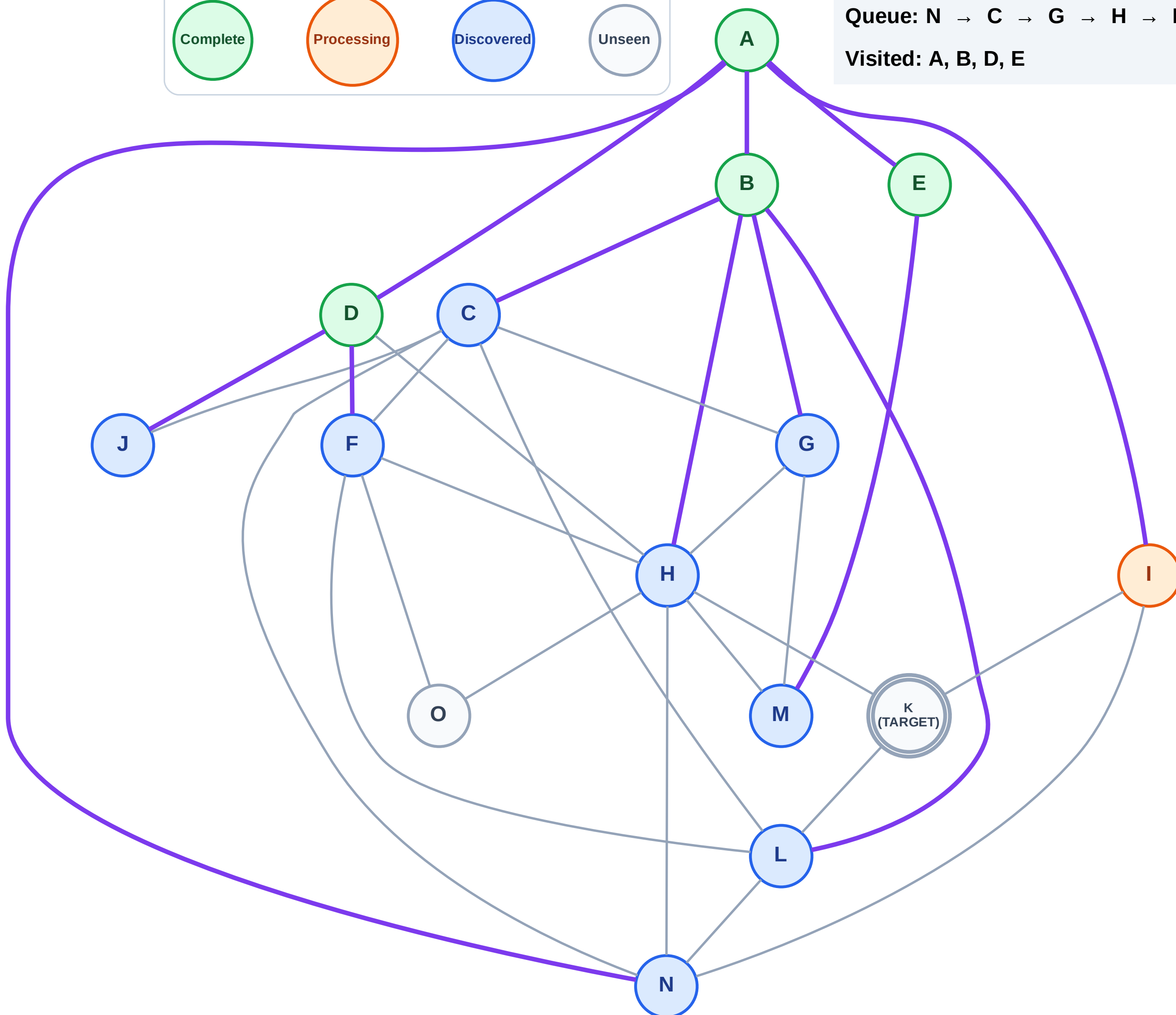
Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle with the label 'Complete', an orange circle with the label 'Processing', a blue circle with the label 'Discovered', and a grey circle with the label 'Unseen'.

Unseen

Visited: A, B, D, E



BFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → I → K

Legend

Complete

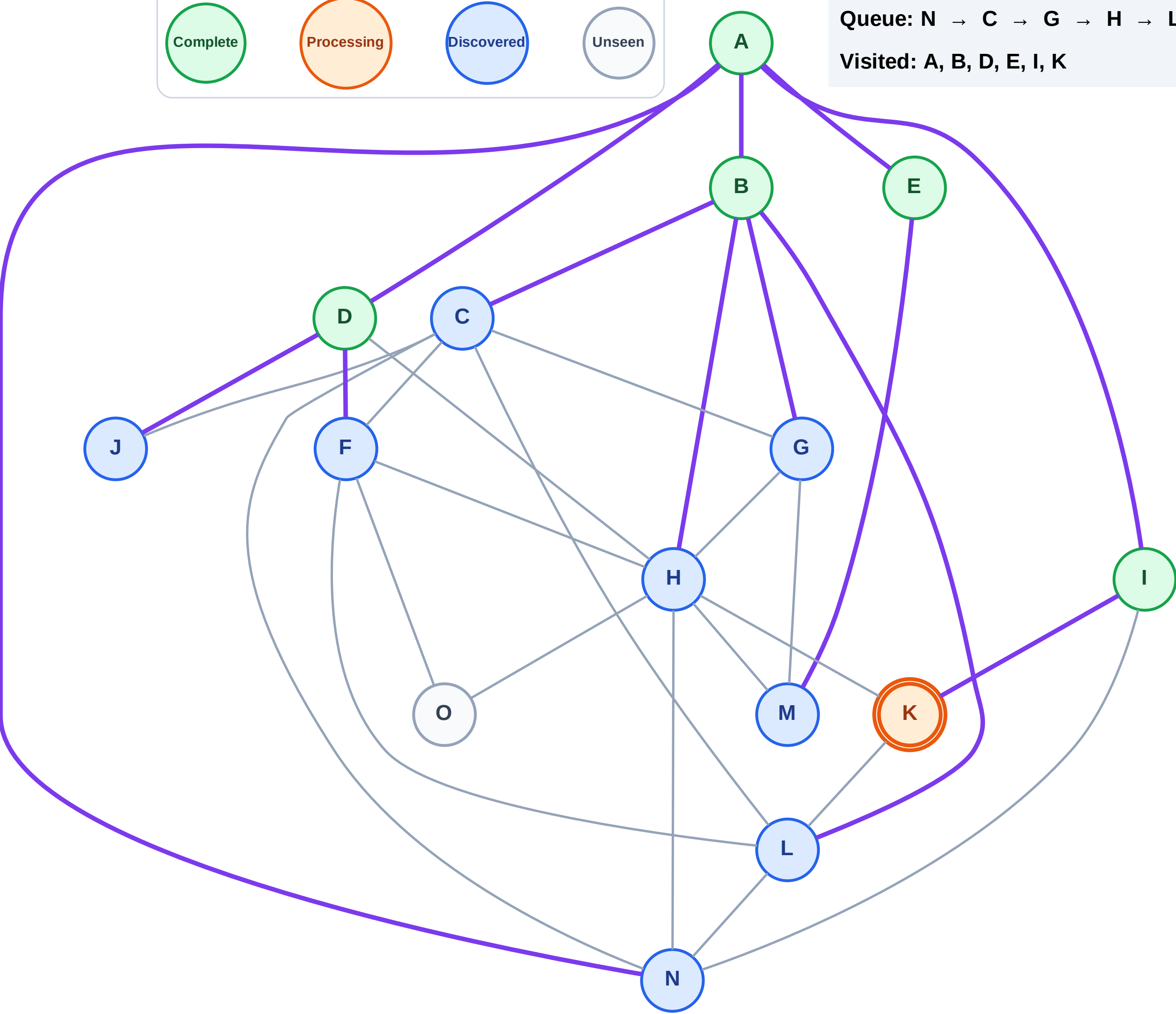
Processing

Discovered

Unseen

Queue: N → C → G → H → L → F → J → M

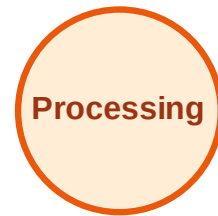
Visited: A, B, D, E, I, K



DFS Traversal

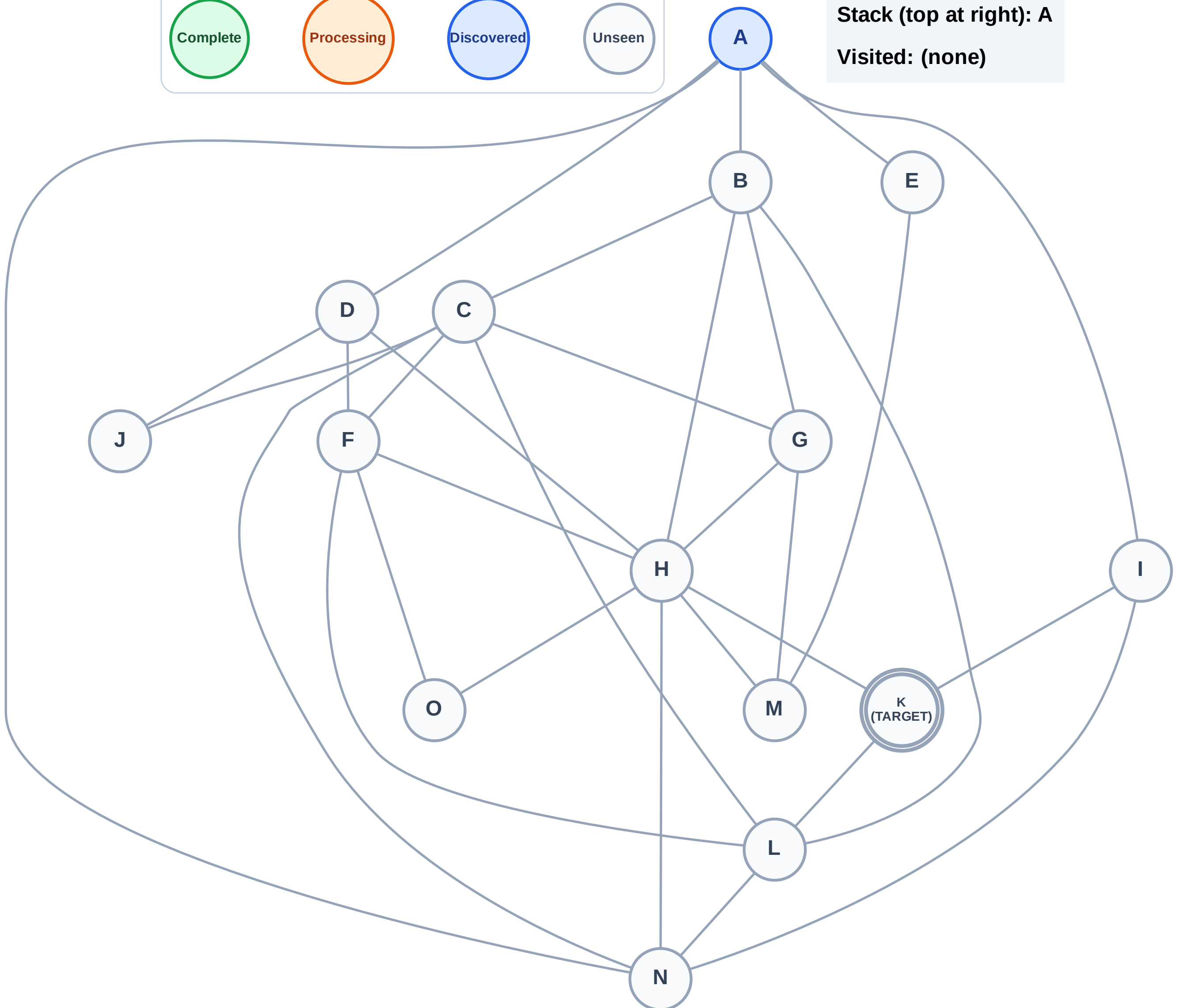
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

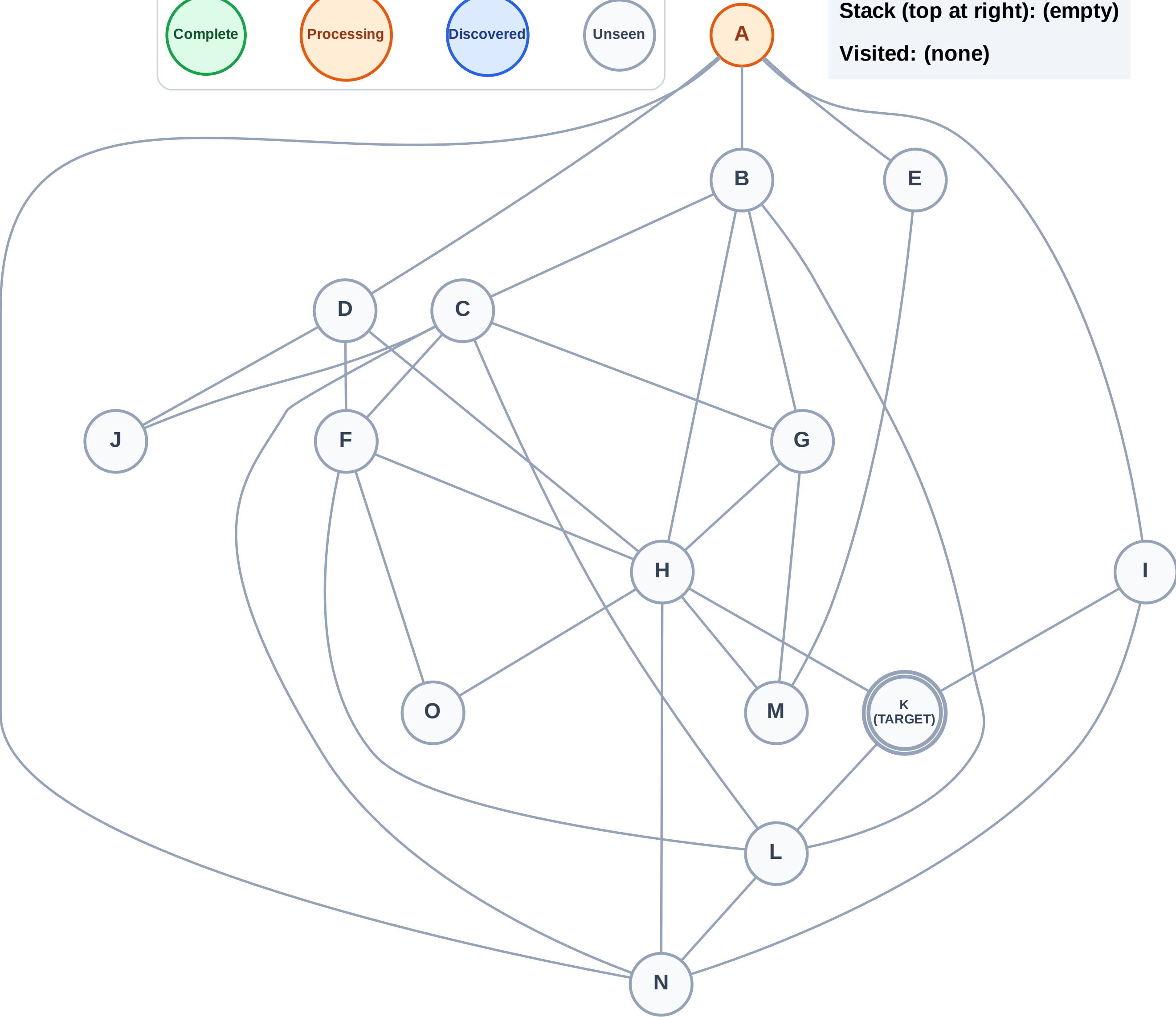
Complete

Processing

Discovered

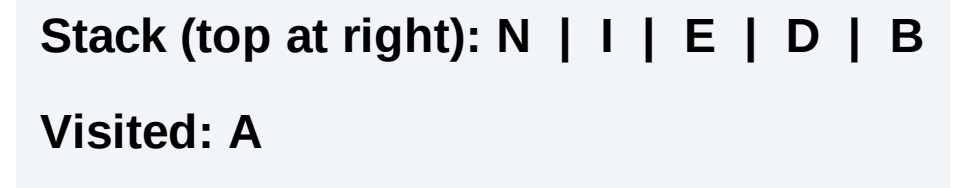
Unseen

Stack (top at right): (empty)
Visited: (none)



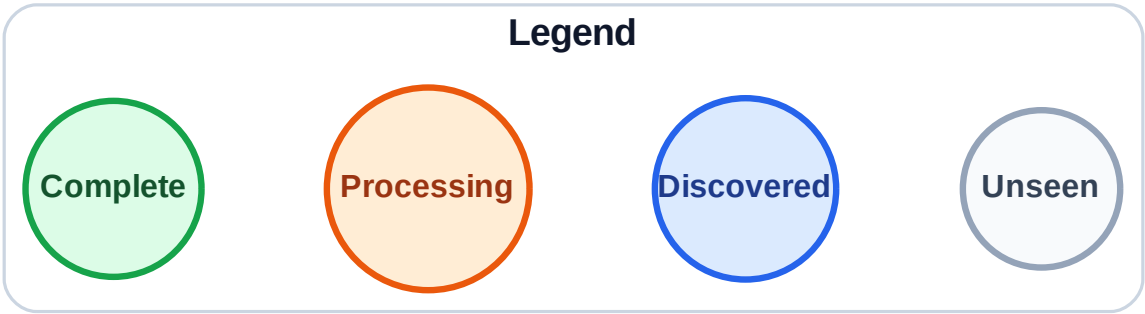
Step 3: Discover N, I, E, D, B from A

Step 3: Discover N, I, E, D, B from A



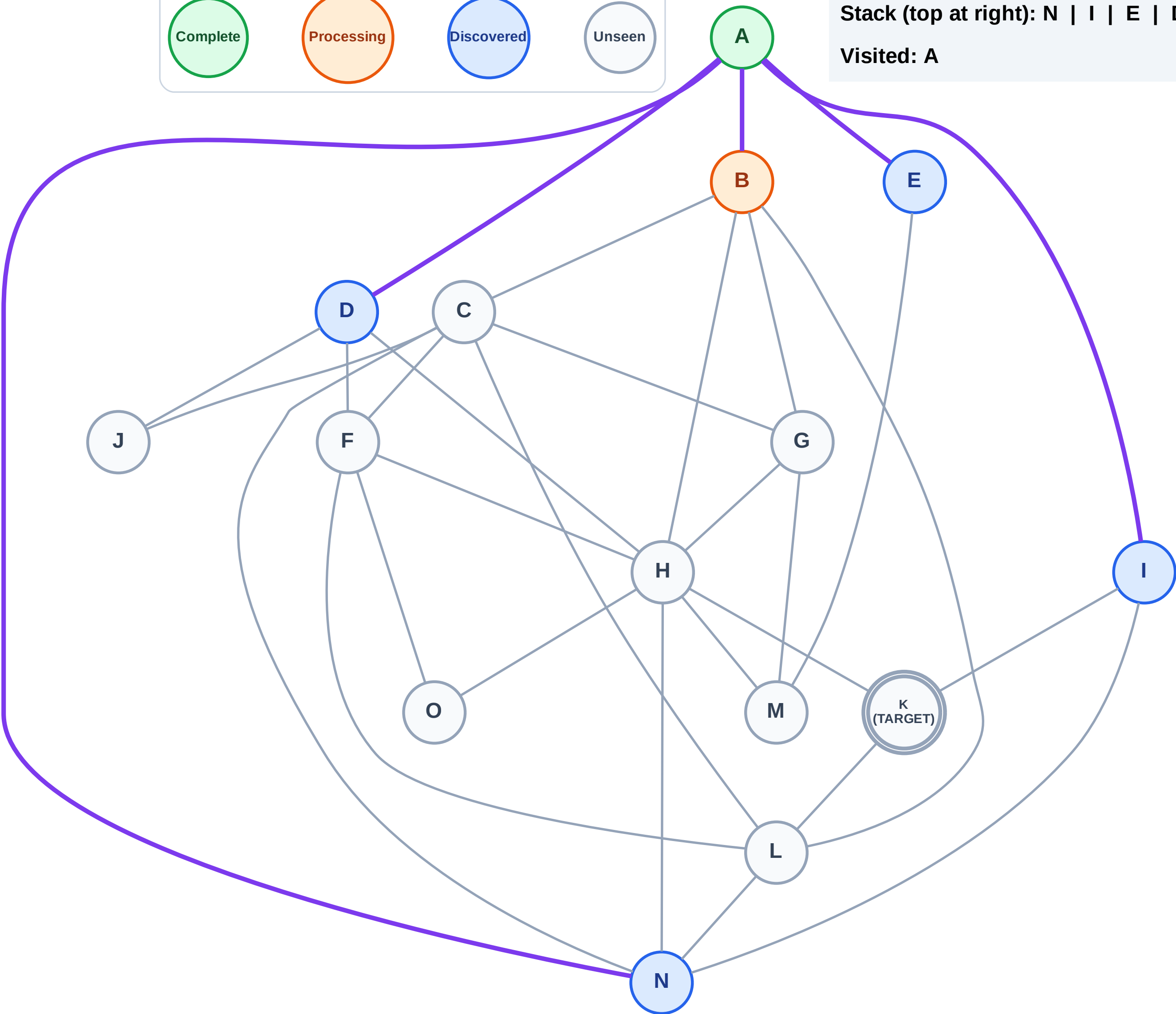
DFS Traversal

Step 4: Process node B



Stack (top at right): N | I | E | D

Visited: A



Step 5: Discover L, H, G, C from B

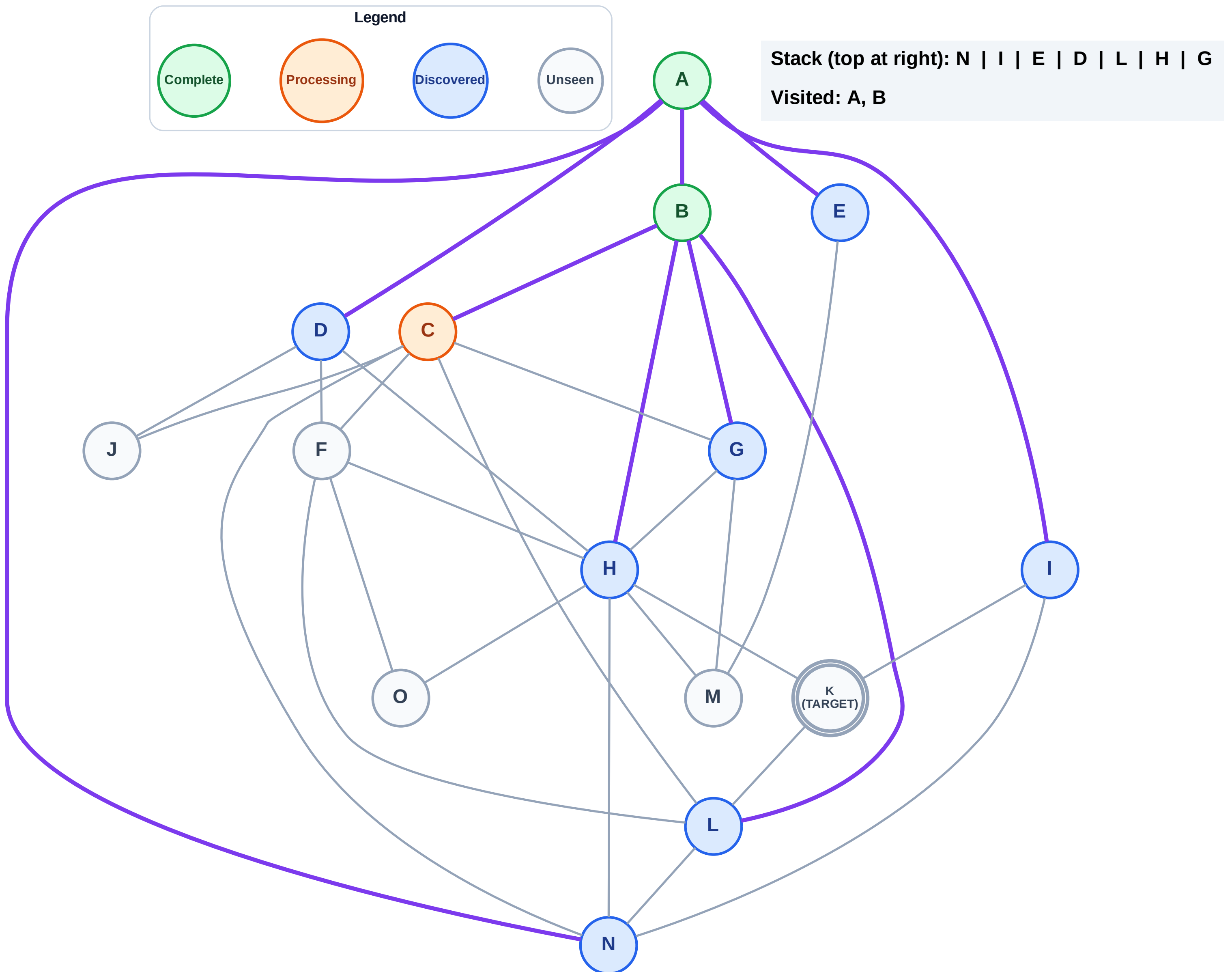
Step 5: Discover L, H, G, C from B



Visited: A, B

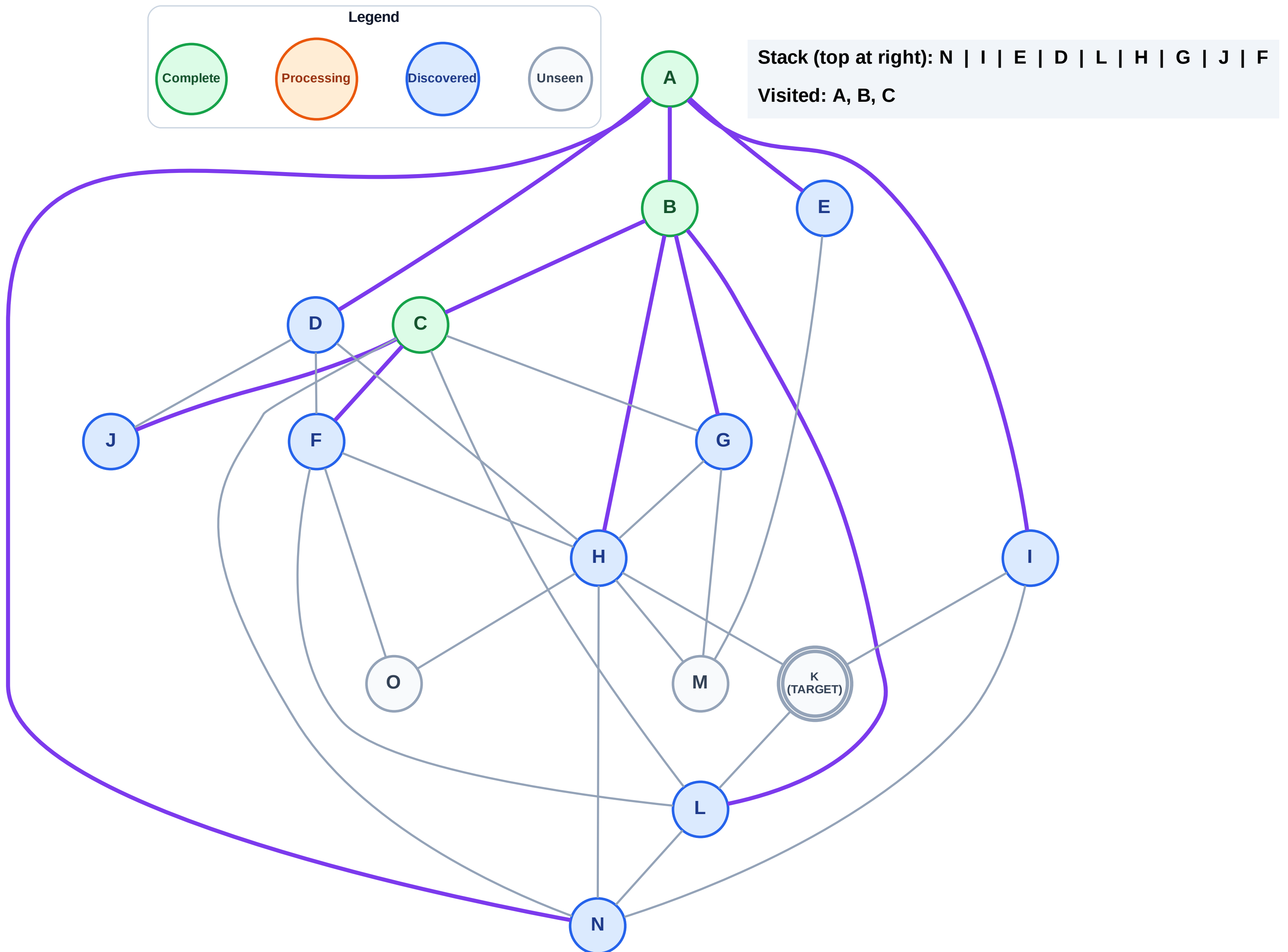
Step 6: Process node C

Step 6: Process node C



Step 7: Discover J, F from C

Step 7: Discover J, F from C



DFS Traversal

Step 8: Process node F

Legend

Complete

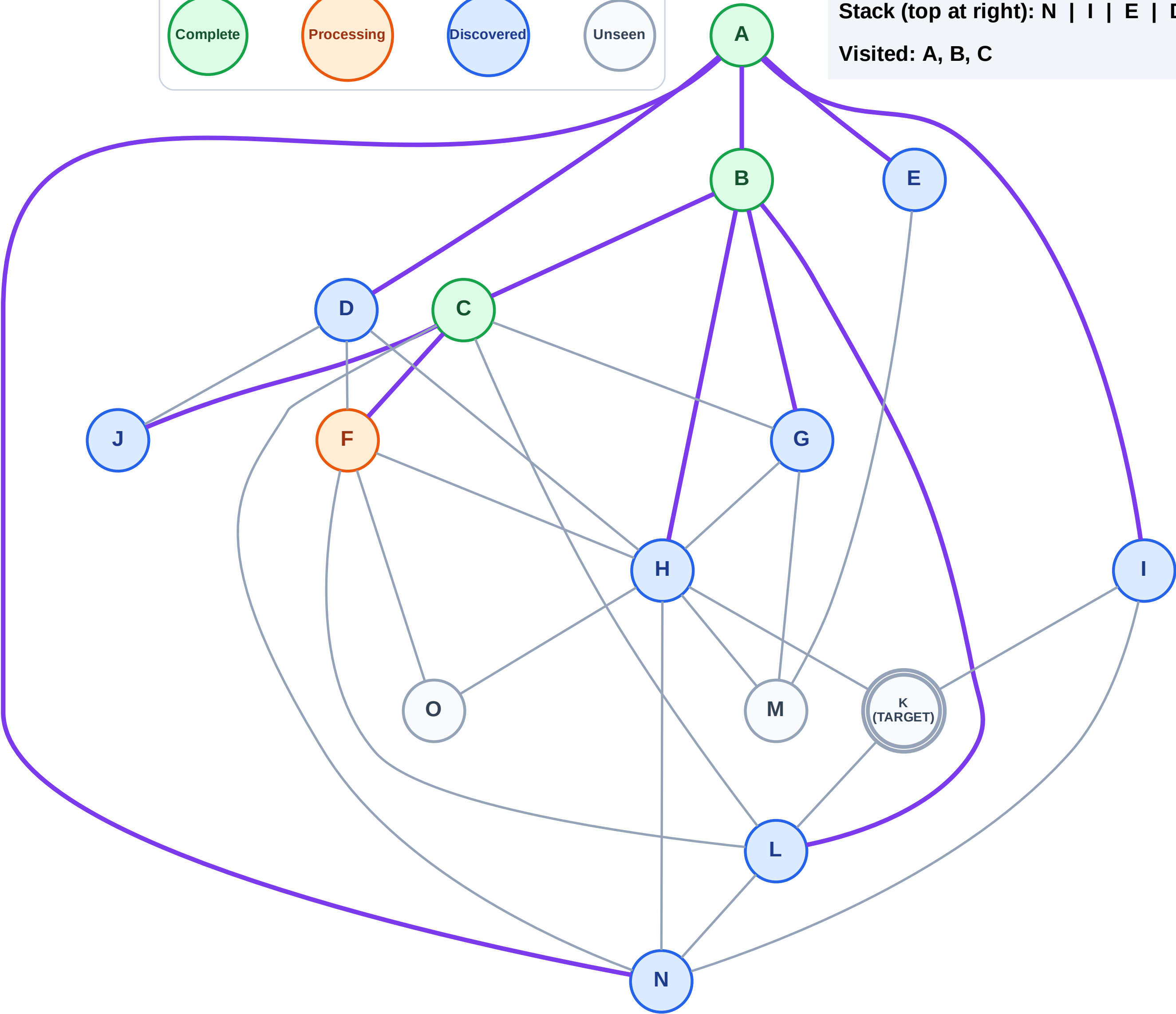
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H | G | J

Visited: A, B, C



DFS Traversal

Step 9: Discover O from F

Legend

Complete

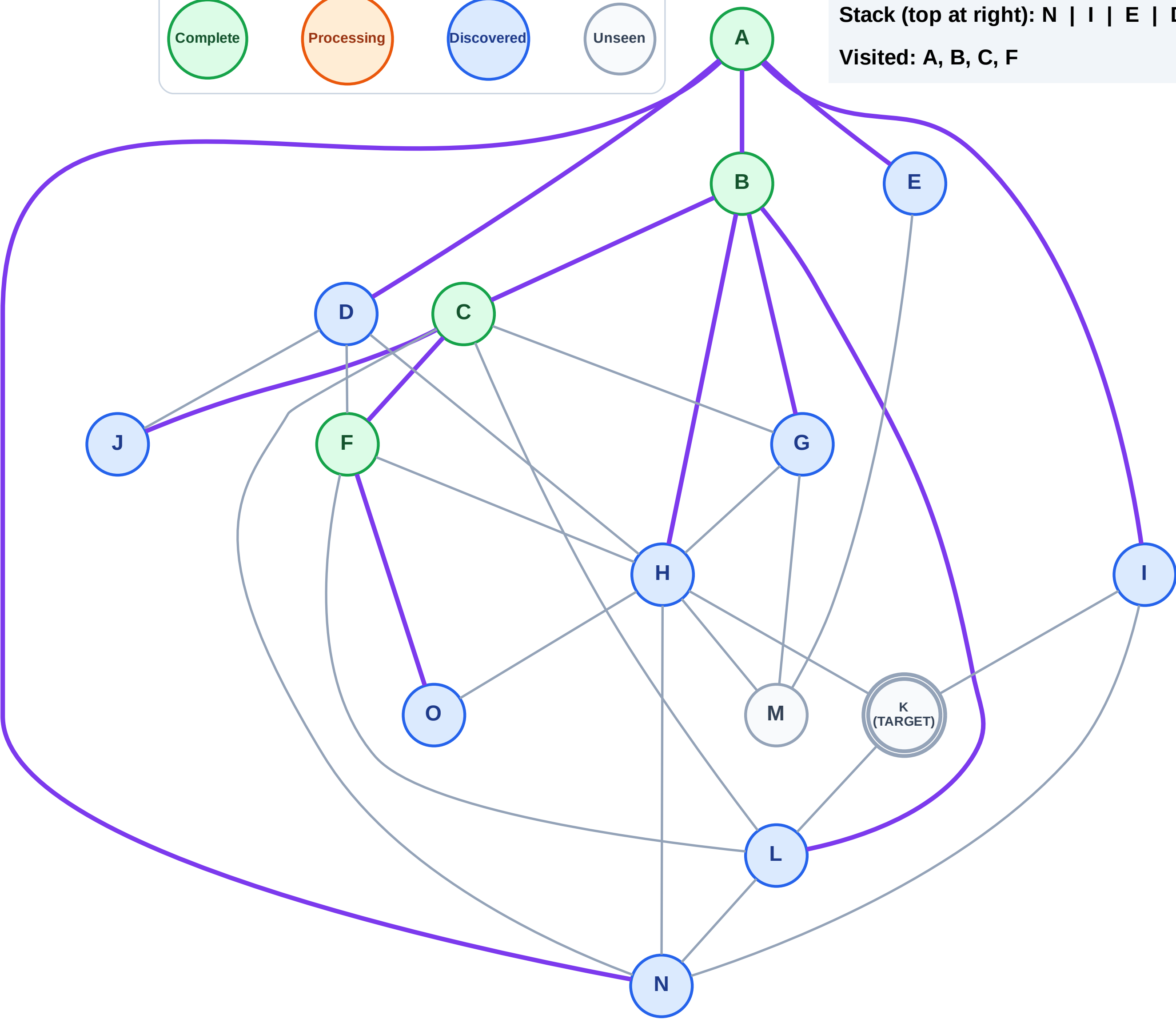
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H | G | J | O

Visited: A, B, C, F



DFS Traversal

Step 10: Process node O

Legend

Complete

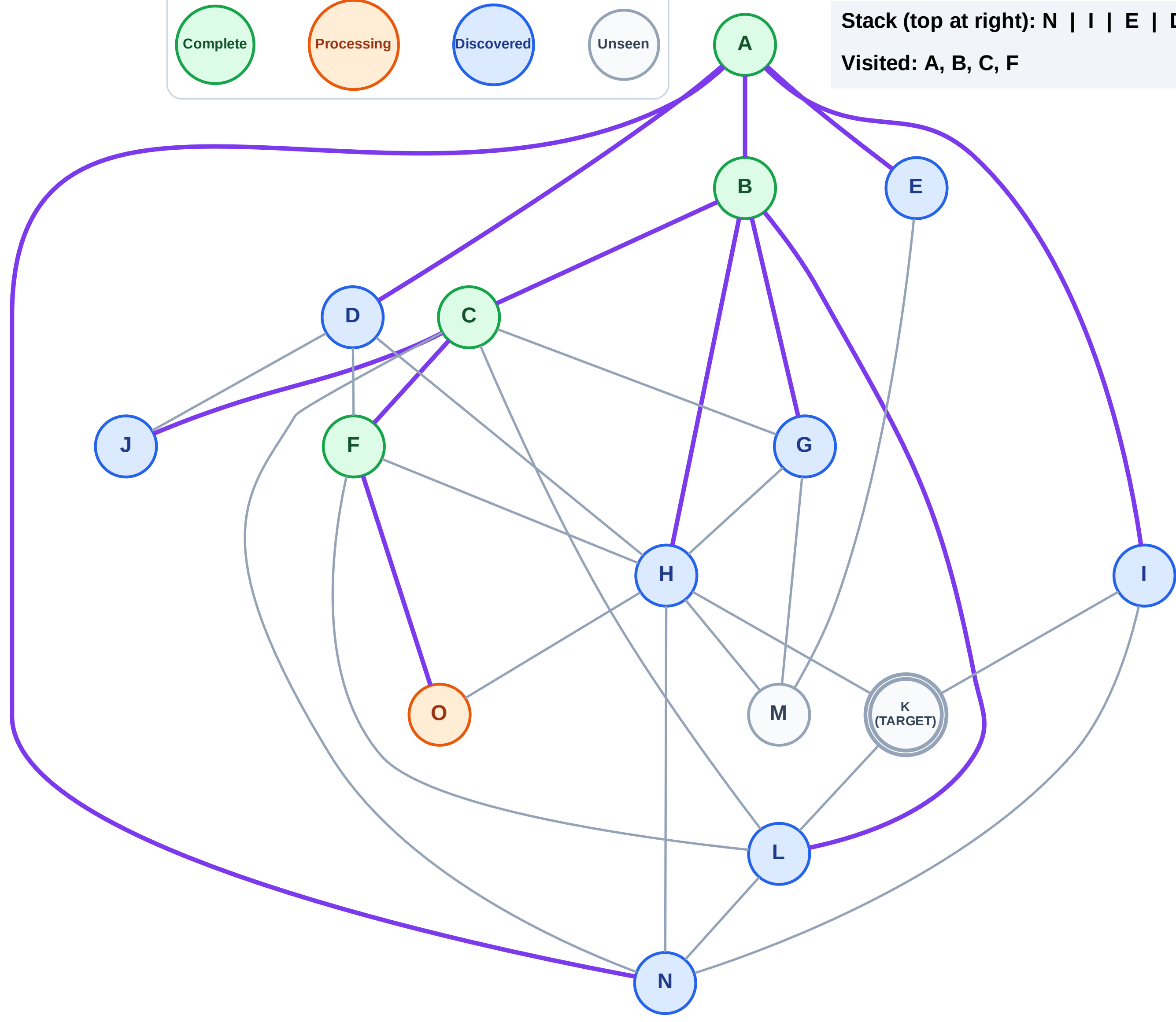
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H | G | J

Visited: A, B, C, F



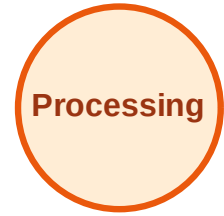
DFS Traversal

Step 11: No new neighbors from O

Legend



Complete



Processing



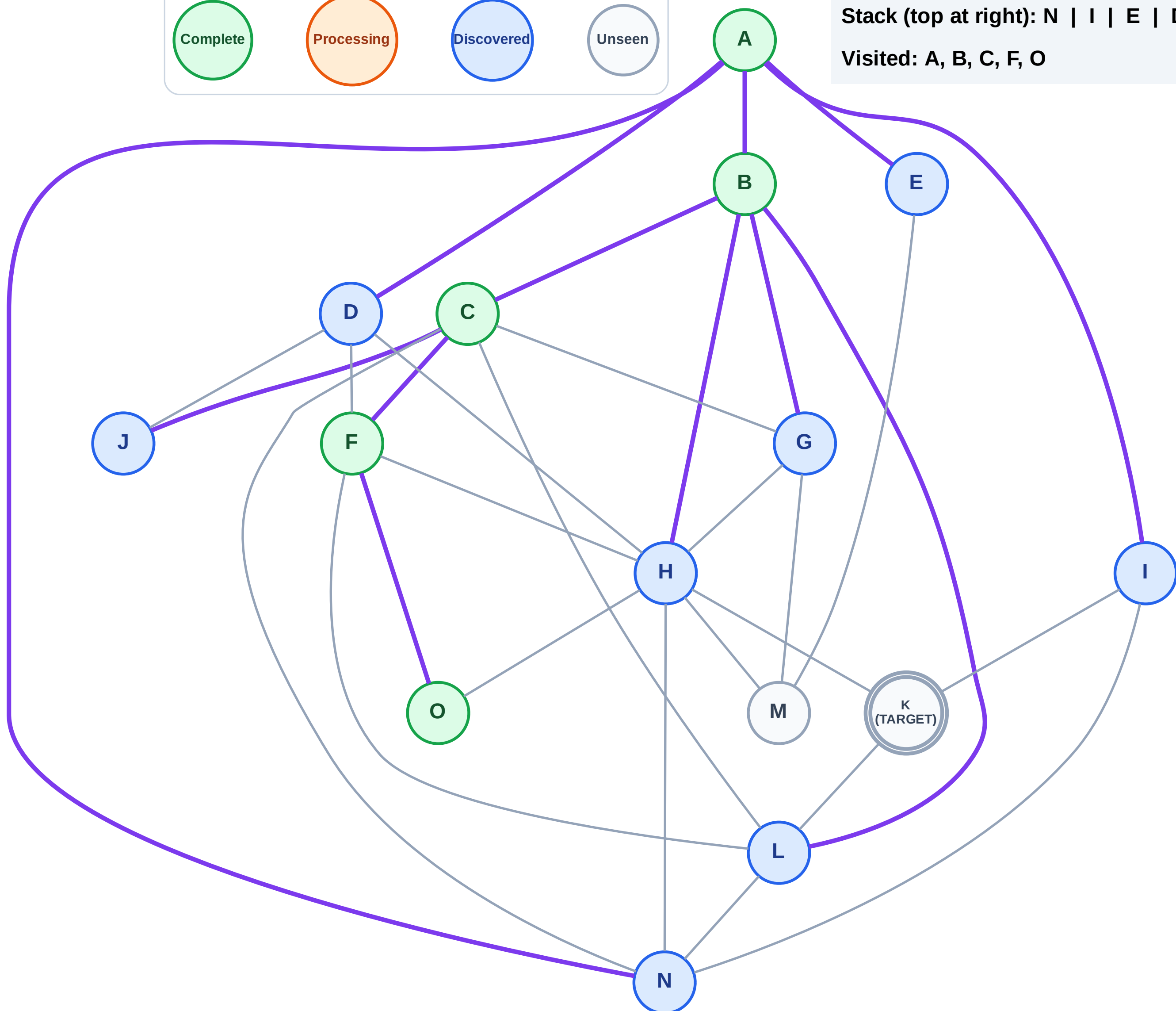
Discovered



Unseen

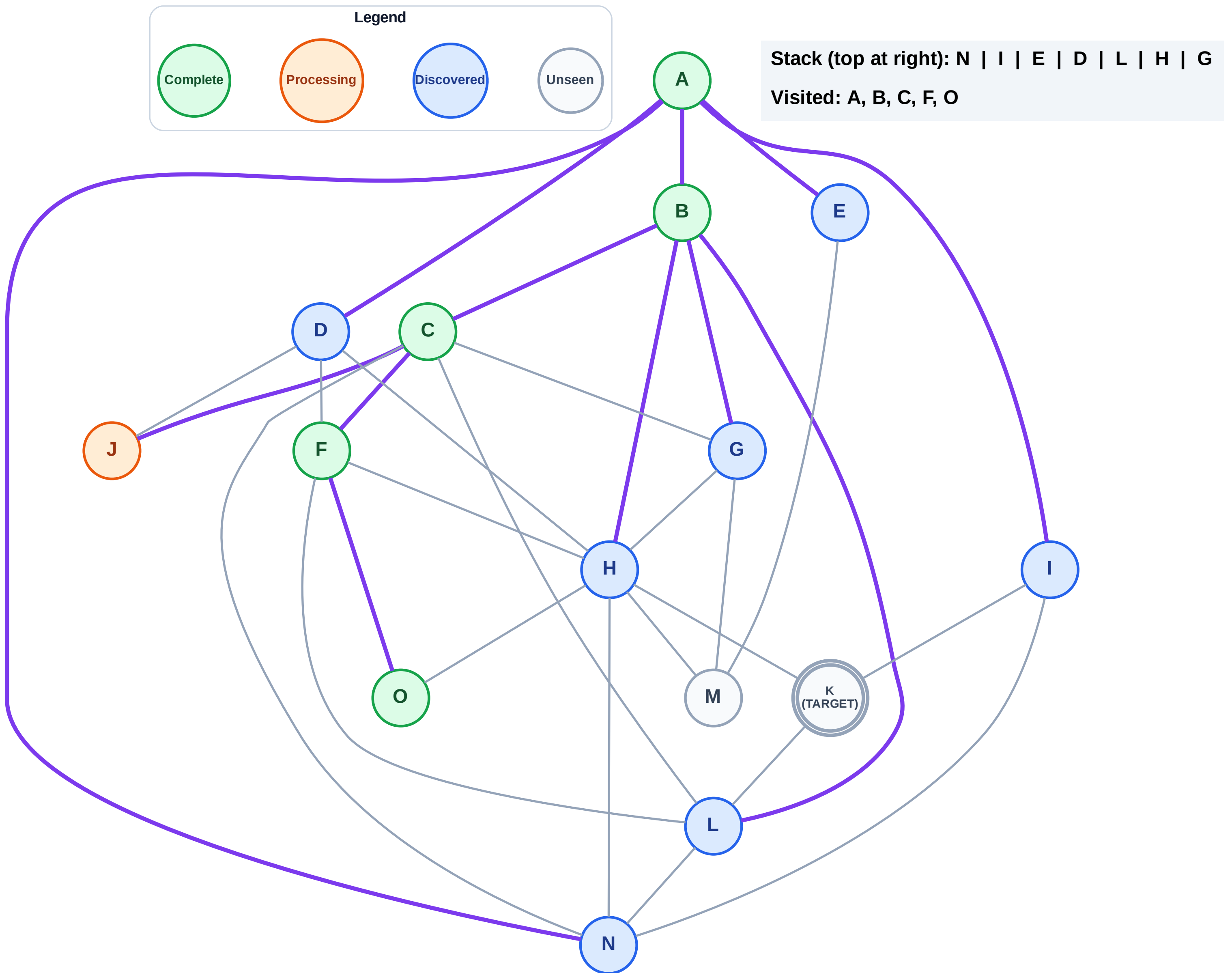
Stack (top at right): N | I | E | D | L | H | G | J

Visited: A, B, C, F, O



Step 12: Process node J

Step 12: Process node J

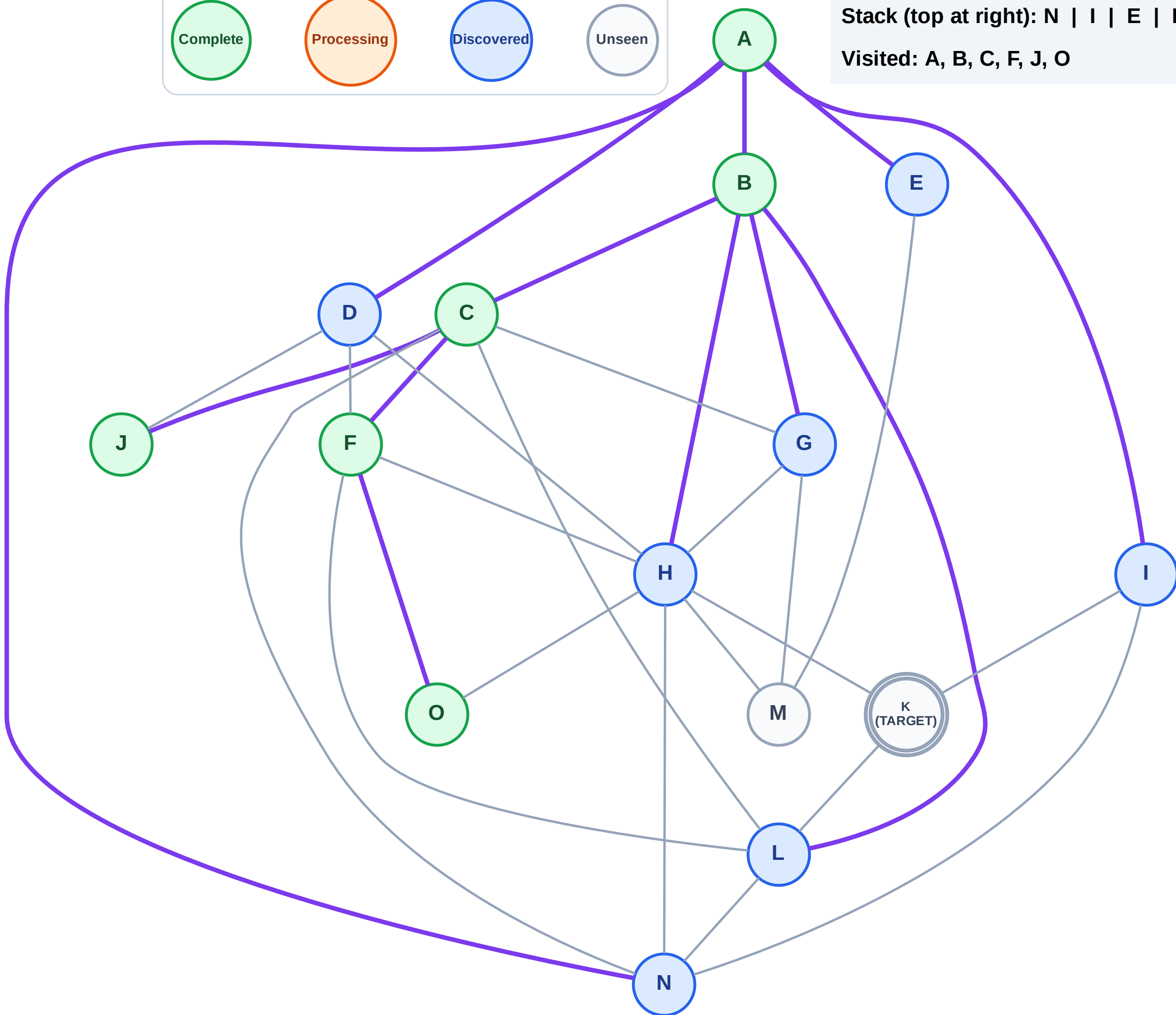


Step 13: No new neighbors from J

Legend

Unseen

Visited: A, B, C, F, J, O



DFS Traversal

Step 14: Process node G

Legend

Complete

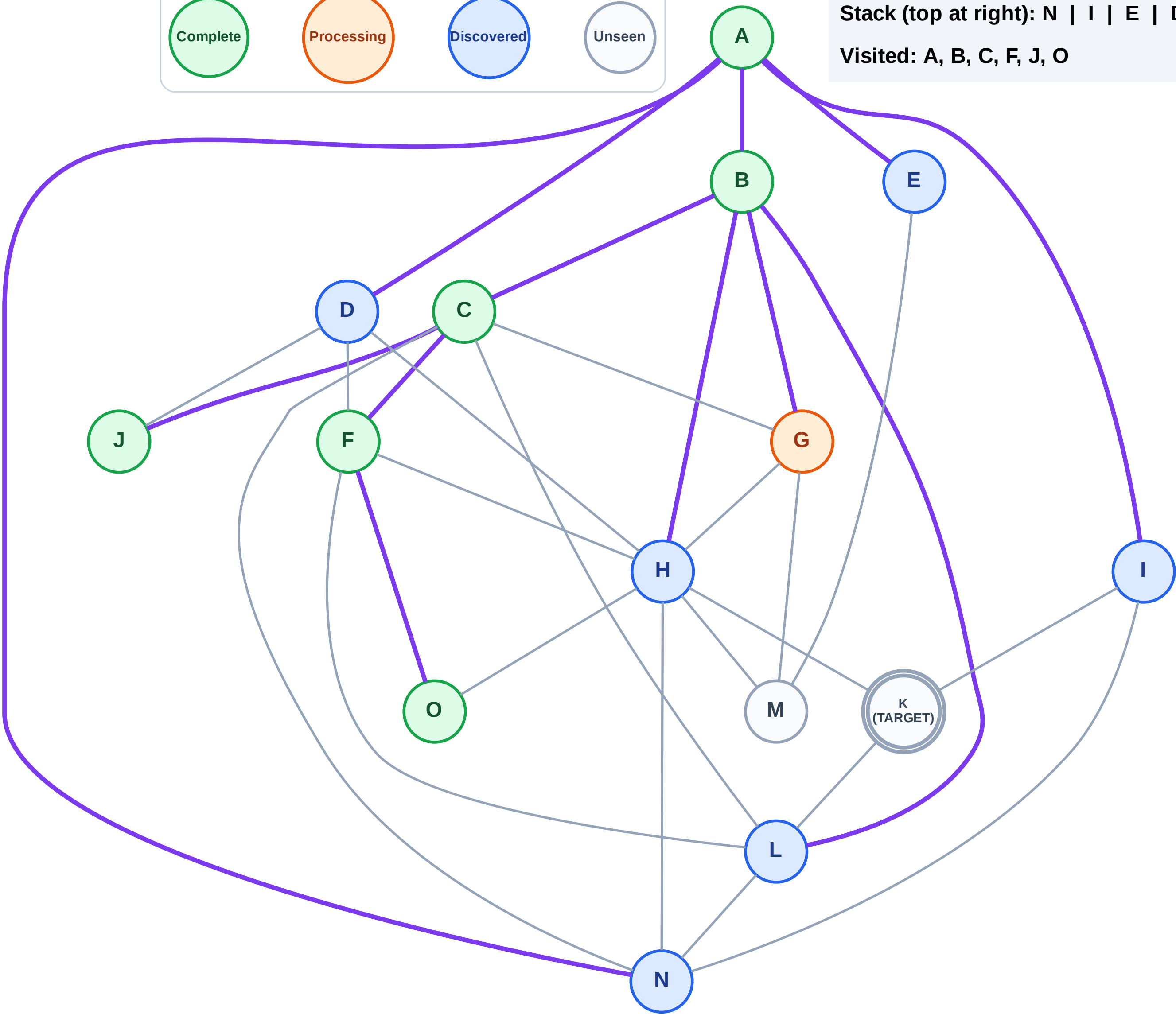
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H

Visited: A, B, C, F, J, O

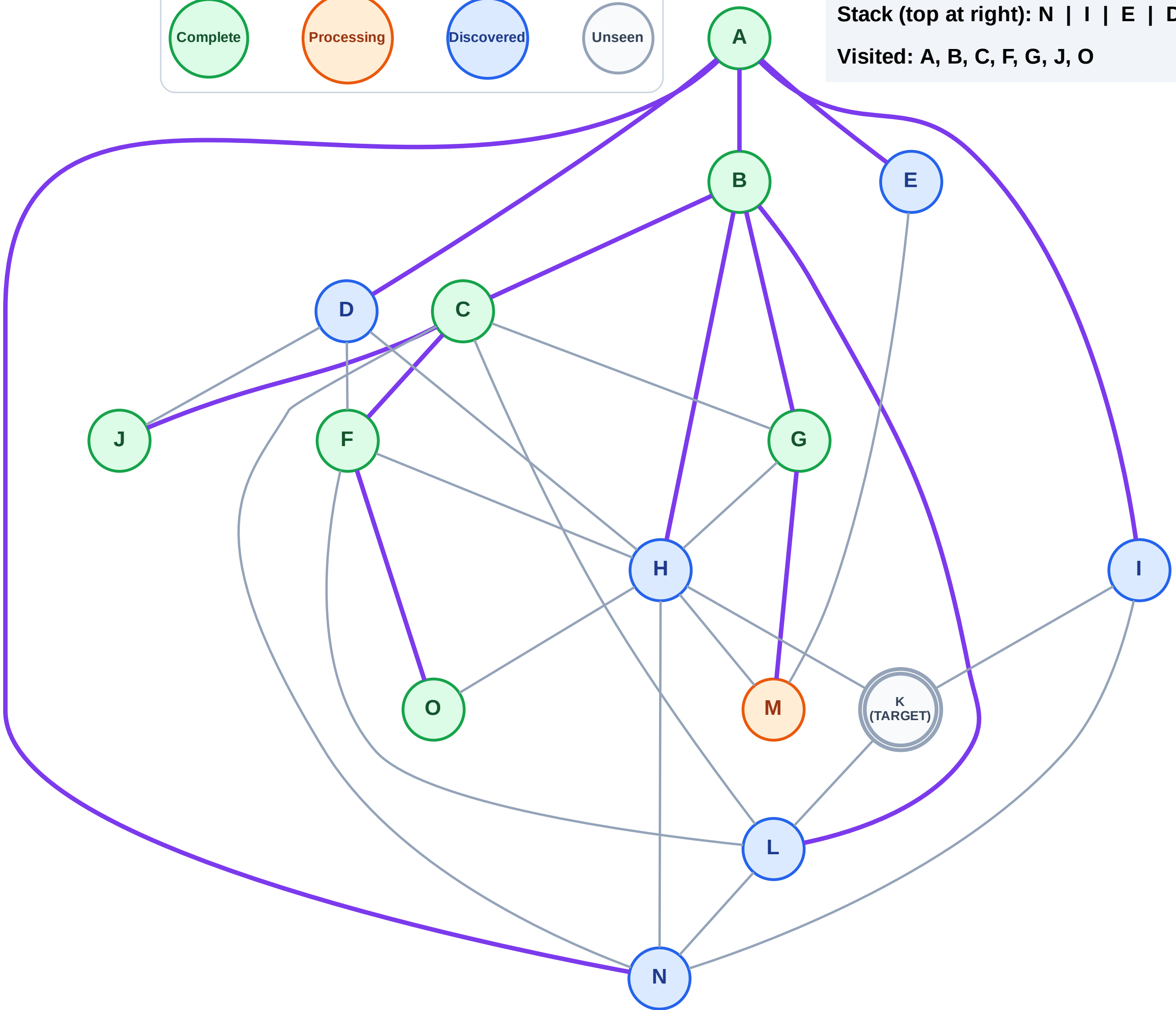


Step 15: Discover M from G

Step 15: Discover M from G



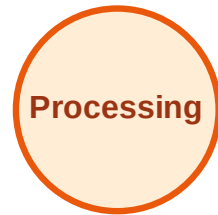
Step 16: Process node M



DFS Traversal

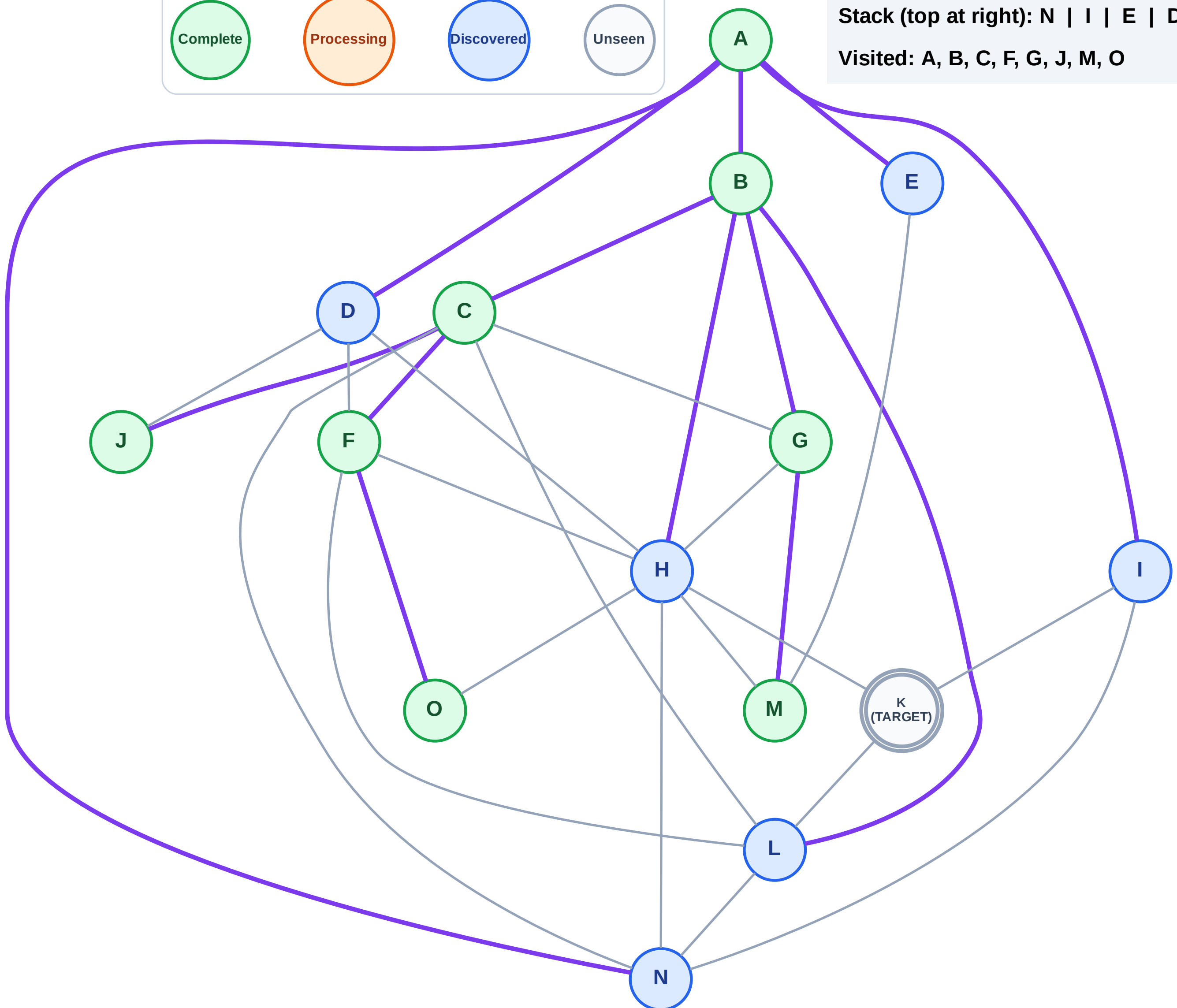
Step 17: No new neighbors from M

Legend



Stack (top at right): N | I | E | D | L | H

Visited: A, B, C, F, G, J, M, O



DFS Traversal

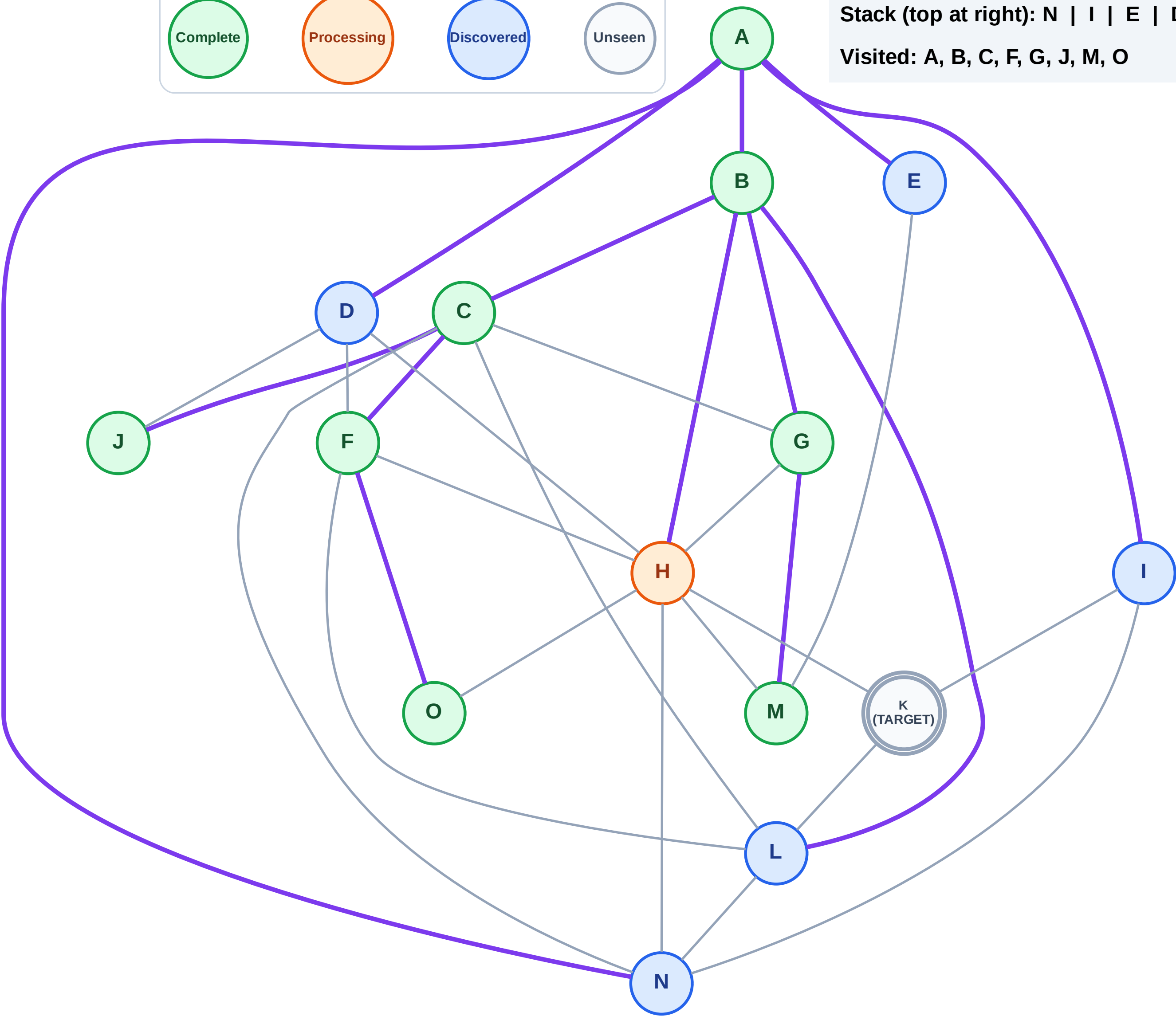
Step 18: Process node H

Legend



Stack (top at right): N | I | E | D | L

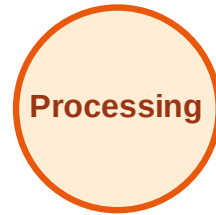
Visited: A, B, C, F, G, J, M, O



DFS Traversal

Step 19: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → H → K

Legend



Stack (top at right): N | I | E | D | L

Visited: A, B, C, F, G, H, J, K, M, O

